

William James Symposium

WILLIAM JAMES ON SENSATION AND PERCEPTION

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Beginning with an analysis of a famous quotation from James' Principles, this essay details some of the highlights of James' treatment of sensation and perception. The primary source is a chapter on discrimination and comparison, in which emphasis is placed on the way people react to difference and sameness, reactions that may be fundamental to the psychophysical methods developed by Fechner and to later techniques, such as stimulus scaling and signal detection procedures, as well as to recent methods (preferential looking and habituation-dishabituation) for investigating perception in human infants. One can see considerable continuity between many of the issues raised in the Principles and those that have been addressed by perceptionists over the ensuing century, but James seems not to have had the direct impact in this domain that he has had, for example, on current conceptions of emotional processes. His associationist and functionalist positions may find contemporary expression in the seemingly disparate approaches known as connectionism and ecological psychology.

In a recent article introducing a special issue of the *Journal of Experimental Psychology: Human Perception and Performance* devoted to research on perceptual development, Eleanor Gibson (1987, p. 515) writes: "Breathes there a psychologist who has never read that 'The baby, assailed by eye, ear, nose, skin and entrails at once, feels it all as one great blooming, buzzing confusion' (James, 1890, I, 488)?" There may, in

fact, be psychologists for whom that quote from James is not familiar. More importantly, how many of those who recognize the quote have actually read the chapter from which it has been extracted? Let me ask those who have not read the chapter, or not read it recently, what do you suppose the chapter is about? Development? Perceptual development? Perception? The last comes closest: The chapter is entitled "Discrimination and Comparison."

What point was James trying to make with his fanciful allusion to "blooming, buzzing confusion?" Typically, that passage from the *Principles* is interpreted to imply that infants have very limited perceptual capacities, that in particular their world is disorganized and chaotic (see, for example, Teller & Movshon, 1986). Gibson, in her article, seems to put special emphasis on the use of the word "assailed," with its implication that infants are passive victims of sensory overload originating in events both within and outside their bodies. Taking the position ("ecological psychology") originally articulated by the late James Gibson (1979), she asserts that, to the contrary, perception is an active process engaged in by curious, searching, moving organisms. Even infants are not helpless in the face of the circumstances that James tried to capture in his famous quote. According to Eleanor Gibson, whose argument is supported by an impressive array of data, infant perception can be characterized as active, exploratory, and motivated, externally directed, dependent on information given in motion, spatially three-dimensional, manifesting constancy, intermodally coordinated, and guiding action in a flexible, rather than mechanical, reflexive, S-R manner (1987, pp. 515-518).

Now, to return to James, I understand his purpose as being not to claim that infant perception is chaotic, but rather to emphasize its initial unity: "one big, blooming, buzzing confusion" (ital-

ics mine). That is, whatever stimuli occur at a given time, regardless of modality, are immediately and automatically fused into an undifferentiated experience. Further, by a process of association, that fusion, and hence "confusion," will remain in place until later stimulation permits it to become segregated into nonarbitrary sub-units through a process of disassociation.

The infant's task, as it matures, is to separate out from this fortuitous aggregation of elementary sensations discrete experiences that, at the very least, accurately map the different modalities (vision, hearing, touch, smell, etc.) and their attributes. The infant, in short, must refine its perceptual discriminations.

One is unlikely to find an extended discussion of discrimination and comparison in current texts, even those written for advanced students and professionals. Nevertheless, I can see, though with no attempt at establishing an actual historical link, two contemporary manifestations of James' preoccupation with discrimination and comparison: (a) the use of perceived differences between stimuli in the service of psychological measurement and (b) the use of natural reactions to sameness and difference as indicator responses in the investigation of memory and perception in nonverbal and preverbal organisms. The latter use brings us, of course, right back to infant perception and is the one I wish to emphasize. But first a bit about threshold measurement and psychophysical scaling.

The classical psychophysical methods and the laws derived from their application (e.g., Weber's Law) were devised for the most part during the last half of the 19th century. James was certainly very much aware of these developments, many of them attributable to Gustav Fechner, and devoted 16 pages to their exposition (I, 533-549), though he found "the proper psychological outcome of that work [to be] just *nothing*"

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1. For brevity, citations of page numbers in *Principles of Psychology*, Volumes I and II, will be given in the form (I, p) or (II, p), for example, (I, 190) for a quotation from p. 190 of Volume I.

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(I, 534). Toward the end of his very competent discussion of this literature, referring to Fechner, James writes:

But it would be terrible if even such a dear old man as this could saddle our Science forever with his patient whimsies, and, in a world so full of more nutritious objects of attention, compel all future students to plough through the difficulties, not only of his own works, but of the still drier ones written in his refutation. (I, 549)

In essence, the classical psychophysical techniques are directed at determining how small a physical change or difference on some dimension can be reliably detected. The experimental method typically involves instructing observers what to look for (for example, Are these two spots of light the same or different in brightness? In which one of those four time intervals did a spot of light appear?) and how to respond (for example, Say "same" or "different." Push one of these four buttons). These methods yield a value (a just-noticeable-difference, a detection threshold) which can then be entered as a dependent measure in equations relating physical variables (for example, the area of a spot of light) to psychophysical performance. Those equations, if sufficiently reliable and generalizable, can take on the status of psychophysical principles or laws.

In a somewhat different use of people's proclivity to react to sameness or difference, various techniques have been created for scaling stimulus magnitude, whether simply on a single dimension (Coombs, 1964; Stevens, 1975) or more complexly in multidimensional space (Shepard, 1974). Such developments in scaling methodology did not directly and explicitly draw on James' insights. But at least one text-book author (Dember 1960) saw the relevance of what James has written about discrimination and comparison for threshold measurement and for the scaling of stimulus magnitude: Same-difference judgments and their variants may have proven so useful as perceptual indicators because responsiveness to sameness or difference (constancy or change) is so fundamentally a property of living organisms. Difference or change is not merely there to be detected; it jumps out at us, grabbing our attention. As James put it:

... There is a real *sensation of difference*, aroused by the shock of transition from one sensation to another which is unlike the first. This sensation of difference has its own peculiar quality, as difference, which remains sensible, no matter of what sort the terms may be, between which it obtains. (I, 495)

Dember saw James' sensation of difference, or as he preferred, this "*reaction to change*" (1960, p. 17), as basic to the psychophysical methods and scaling techniques. That point becomes especially cogent when one considers the methods recently developed for investigating perception in human infants.

In conventional psychophysical research, the subject (observer) is instructed what to look for and how to respond, procedures obviously not available for use with animals or human infants. It turns out that James' "shock of difference" is even a more pertinent basis of methodology for use with preverbal or nonverbal subjects than it is with the linguistically competent. One simply capitalizes on the organism's natural response to change, or to difference, as an indicator response of perceptual sensitivity. Operant techniques are also available, with reinforcement contingencies serving as the medium of communication between experimenter and subject and operant responses as the subject's way of indicating perceptual discrimination. But it is the methodology employing spontaneous, untrained responses that James presaged in the passage above. Thus, for example, an infant will look at a particular visual stimulus, say a triangle, that is placed in its field of view. But if the triangle remains unchanged for some time, the infant's interest will wane and "looking time" will decline, reflecting a process commonly referred to as habituation. However, if a new figure, say a square, replaces the triangle, looking time will increase (dishabituation), but only, of course, if the triangle and square are discriminable. Using techniques based on that simple logic, psychologists have uncovered a wealth of information about the visual and other sensory capacities of human infants and have been able to trace the time course of their development and pinpoint the factors that facilitate or hinder that development (Aslin, 1987; Siegler, 1986). Another fruitful procedure, the "preferential looking technique," based on in-

fant's natural preferences for certain visual patterns over other, has also been extensively employed (Teller and Movshon, 1986). Preferential looking may not derive from exactly the same underlying processes as habituation-dishabituation, and may not seem to fit James' "shock of transition" as closely; however, it may be that what underlies preference is differential habituation: Some patterns take longer to habituate than others and hence are examined for longer periods of time. In this regard, there are now strong indications that how long an infant takes to habituate to a variety of visual patterns is predictive of later intellectual functioning: The quicker the habituation, the brighter the child (Fagan & Montie, 1988). These data provide the first solid evidence of cognitive continuity between infancy and later childhood in a normal population.

So much for James' "blooming, buzzing confusion." What other links might there be between topics in sensation and perception addressed in the *Principles* and current concerns? It should come as no surprise that the links are many. Perceptionists do not easily let go of problems, perhaps because those problems remain recalcitrant to ultimate solution despite increasingly clever models and methodologies. For example, the classical psychophysical methods, though still in use, have largely given way to the much more sophisticated procedures for data collection and analysis introduced by Wilson Tanner, John Swets, and David Green in the late 1950s (see, for example, Green & Swets, 1964). With regard to content, as contrasted with technique, James devoted several chapters to perceptual issues; though the volume of research on those specific topics may have waxed and waned over the past century, it is fair to say that they remain central issues in textbooks and handbooks of perception, and many are still being investigated in the laboratory.

For example, in a chapter entitled "The Perception of Things," James devoted considerable attention to visual illusions. As perceptionists are wont to do, James saw illusions as more than curiosities: They reveal or highlight certain psychological principles. With regard to illusions, James' own theoretical stance seems much akin to that of Helmholtz in his time and to that of the "transactional

functionalists" of the 1940s and '50s (see, for example, Ittelson & Kilpatrick, 1951), attributing various illusions to inferences drawn on the basis of prior experience, or associative processes. The moon illusion (the increase in the apparent size of the moon when it lies close to the horizon), for example, is explained by James by reference to certain cues which render the moon apparently more distant than "when it rides on high" (II, 92). Since there is no change in the moon's visual angle between those extreme positions, "we deem it must be a larger body, and we so perceive it" (II, 93). This account continues to be offered by textbook writers, although serious students of the moon illusion (see, for example, Hershenson, 1989) find it unsatisfactory for various reasons, one of which is readily apparent to any moon gazer: The horizon moon actually seems closer, as well as larger, than the zenith moon. More generally, one may ask how the inferential processes, summarized under the seductive phrase, "we deem that," actually work. In any case, James treated a set of such illusions (those of "the first type") as all following the same general principle: "when exposed to ambiguous stimulation, we see the most probable object" (II, 95).

In James' "illusions of the first type," one perceives the most probable object (e.g., a larger than normal moon) given ambiguous stimulation in combination with associations formed through prior experience. His "illusions of the second type," from the present vantage point, seem hardly any different. In contemporary terminology, we would attribute them to expectancies. As James puts it, we perceive "the wrong object" because "the mind is temporarily full of the thought of the object" (II, 86). He goes on to cite numerous anecdotes, some from his own experience, which illustrate how expectancy can affect perception, as, for example, in what he calls the proof-reader's illusion:

I remember one night in Boston, whilst waiting for a 'Mount Auburn' car to bring me to Cambridge, reading most distinctly that name upon the signboard of a car on which (as I afterward learned) 'North Avenue' was painted. (II, 96)

Considerable experimental support

for the proposition that expectancies can influence perception can be found in the vast literature of the 1940s and 50s falling under the rubric of the "New Look in Perception" (see for example, Dember & Warm, 1979, chapter 9) and in the more recent work on "priming" (e.g., Marcel, 1983). While in current usage the effects on perception of cognitive and affective variables are typically not referred to as illusory, the phenomena certainly fit closely James' summary principle, which he alludes to as "the general law of perception," to wit: "*whilst part of what we perceive comes through our senses from the object before us, another part (and it may be the larger part) always comes . . . out of our own head*" (II, 103).

As noted, this discussion of illusions is to be found in James' chapter on "The Perception of Things." The theoretical framework is quite explicitly cognitive. But James also treated earlier in the text, in a chapter on "Sensation," phenomena akin to illusions that he attributed to physiological, as distinct from associative, processes. Successive and simultaneous color and brightness contrast, discussed at great length (II, 13-27), provide examples of such phenomena. In these instances, James departs from his Helmholtzian view in asserting that contrast phenomena are sensory rather than perceptual, arising directly from "*physiological processes, and not deception of judgment*" (II, 26). It is interesting to note that James arrives at this conclusion by way of a critical analysis of the research methodology employed by proponents of the conflicting positions. Clearly, James was prepared to draw on and carefully evaluate laboratory research where available and did not rely solely on anecdote or armchair speculation, as may seem to be the case.

Space and time limitations preclude a detailed discussion of the several additional topics pertinent to this article that James covers in the *Principles*. To two of them, James devoted an entire chapter each. Brief mention of the highlights of those chapters should suffice to give a good sense of the extent to which current concerns mesh with those of James.

The chapter on time perception (I, chapter XV), James indicates in a footnote, was "reprinted almost verbatim from the Journal of Speculative Philos-

ophy." Even so, it is replete with data and generalizations that have withstood the test of time. For example, "*Our sense of time, like other senses, seems subject to the law of contrast.*" "Like other senses, too, *our sense of time is sharpened by practice.*" "*Tracts of time filled (with clicks of sound) seem longer than vacant ones*" (All quotes from I, 618). "*The same space of time seems shorter as we grow older*" (I, 625). While time perception is not high on the list of contemporary "hot topics," it nevertheless has been regularly investigated over the past century (see, for example, Doob, 1971; Fraisse, 1964, 1984). I would also include, in recent work on time perception, studies of the "psychological moment" (investigated through judgments of simultaneity and successivity (treated extensively by James) and elegantly modeled by Alfred Kristofferson (1984). Finally, I want to mention one aspect of time perception that seems to have escaped James' notice: The perception of temporal patterns in music, a topic that only very recently has come under laboratory examination (see, for example, Jones, 1989).

By far the longest chapter (148 pages) in both volumes of the *Principles* is that on Space Perception. With larger type and margins, more illustrations, "boxes," practice test questions, exercises, references, and indexes, it could be a book in itself by modern standards. To give a sense of the topics covered, I will have to resort to a partial listing and confine myself to global categories at that. In this ambitious chapter, James discusses, among other things, the perception of extensiveness and voluminousness; the perception of spatial order and spatial relations (left-right, up-down); the sensation of motion (as distinct from *motion perception*, an inferential process) and illusory motion; kinesis; space perception by the blind; visual depth perception (via retinal disparity as well as convergence, accommodation and various monocular cues); illusions of depth. Throughout, there is an effort to show how elementary sensations are transformed into percepts through associative and cognitive processes. The Gestaltists' nativist position, as exemplified in the principles of organization, had not yet been well-articulated, but the fundamental issue they

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dwelled on—whence organization in perception?—was certainly of deep concern to James, though he happened to choose an associationist answer.

Of the topics discussed in the chapter on the perception of space, perhaps the most enduring have been motion and depth, with the latter greatly illuminated by Béla Julesz's (1971) introduction of random dot stereograms and subsequently with the widespread availability of color video systems for the rapid generation of very complex visual displays.

What did James contribute to our understanding of sensation and perception? Remember that the *Principles* was meant to be a textbook for a course in general psychology. As such, its main function was to evaluate and communicate to students what others had found or proposed. But, unlike most modern textbooks, the *Principles* clearly and explicitly reveals its author's own experiences and theoretical biases and hence offers much more than careful and judicious reporting. Nevertheless, it is hard to see on the surface any more in James' treatment of sensation and perception than intelligent and well-informed commentary. Aside from the "blooming, buzzing confusion" with which this article began, contemporary writers and researchers on the topics of sensation and perception rarely cite James' ideas or even terminology as seminal for their own work—by contrast with his still very much alive-and-well assertions about

emotion. Not that his S & P chapters do not bear re-reading; on the contrary, they are filled with insights and examples both from "real life" and from the laboratory that at the very least can help students and researchers locate their narrow, parochial problems in a richer, broader psychological/philosophical context. And, without stretching too far, we might see in James' dual emphases on associationism and functionalism harbingers of two of the most prominent and promising current approaches to learning and perception: connectionism (e.g., McClelland & Rumelhart, 1986) and ecological psychology (e.g., Gibson, 1979), respectively. Perhaps in the next century, a new William James will find a way of synthesizing those two powerful, but seemingly disparate points of view.

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